Southern Sweetlips: extra slides

Mark Bravington, CSIRO: June 2021

Dealing with length in age-based CKMR

Age-based pop dyn much simpler than joint-L-A

but *length* drives fec and sel

Ideal: known L & A for each adult

Need to back- (POP) and fwd- (HSP) project L for ***E****RRO*

Dealing with length in age-based CKMR

Age-based pop dyn much simpler than joint-L-A

but *length* drives fec and sel

Ideal: known L & A for each adult

Need to back- (POP) and fwd- (HSP) project L for ***E**RRO*

My suggestion: *"on average, you stick to your quantile"*

- extra easy If CV(L@A) is constant: indiv Linf, indiv vonB
- not hard anyway

For HSP calcs: average (fec^2) is **not** (average fec)^2

- instead, pick 5 (or 10) quantiles within L@A, project from each

Dealing with length in age-based CKMR

Age-based pop dyn much simpler than joint-L-A

but *length* drives fec and sel

Ideal: known L & A for each adult

Need to back- (POP) and fwd- (HSP) project L for ***E****RRO*

My suggestion: *"on average, you stick to your quantile"*

- extra easy If CV(L@A) is constant: indiv Linf, indiv vonB
- not hard anyway

For HSP calcs: average (fec^2) is **not** (average fec)^2

- instead, pick 5 (or 10) quantiles within L@A, project from each

If sel-within-age strong enough to affect Z-within-age... yuk!

Tedia #1: adult length but not age

We just calculated

$$\mathbb{P}\left[\mathrm{xxP} \middle| b_{\mathrm{Julie}} a_{\mathrm{Adam}} \ell_{\mathrm{Adam}} y_{\mathrm{Adam}}\right]$$

But don't know a_{Adam}

Tedia #1: adult length but not age

We just calculated

$$\mathbb{P}\left[\mathrm{xxP}|b_{\mathrm{Julie}}a_{\mathrm{Adam}}\ell_{\mathrm{Adam}}y_{\mathrm{Adam}}\right]$$

But don't know a_{Adam}

So we need to work out

$$\mathbb{P}\left[\mathrm{xxP}|b_{\mathrm{J}}\ell_{\mathrm{A}}y_{\mathrm{A}}\right]$$

$$= \sum_{a^*}\mathbb{P}\left[\mathrm{xxP}|b_{\mathrm{J}}, a_{\mathrm{A}} = a^*, \ell_{\mathrm{A}}y_{\mathrm{A}}\right] \times \mathbb{P}\left[a^*|\ell_A y_A\right]$$

Tedia #1: adult length but not age

We just calculated $\mathbb{P}\left[\text{xxP}|b_{\text{Julie}}a_{\text{Adam}}\ell_{\text{Adam}}y_{\text{Adam}}\right]$

But don't know a_{Adam}

So we need to work out

$$\mathbb{P}\left[\text{xxP}|b_{\text{J}}\ell_{\text{A}}y_{\text{A}}\right]$$
$$= \sum_{a^*} \mathbb{P}\left[\text{xxP}|b_{\text{J}}, a_{\text{A}} = a^*, \ell_{\text{A}}y_{\text{A}}\right] \times \mathbb{P}\left[a^*|\ell_A y_A\right]$$

... so we need $\mathbb{P}\left[a|\ell y\right]$

Tedia #1: adult length but not age

We just calculated $\mathbb{P}\left[\text{xxP}|b_{\text{Julie}}a_{\text{Adam}}\ell_{\text{Adam}}y_{\text{Adam}}\right]$

But don't know a_{Adam}

So we need to work out $\mathbb{P}\left[\text{xxP}|b_{\text{J}}\ell_{\text{A}}y_{\text{A}}\right]$

$$= \sum_{a^*} \mathbb{P}\left[\text{xxP}|b_{\text{J}}, a_{\text{A}} = a^*, \ell_{\text{A}}y_{\text{A}}\right] \times \mathbb{P}\left[a^*|\ell_A y_A\right]$$

... so we need $\mathbb{P}\left[a|\ell y\right]$

Direct A@L samples? IMO the only *good* "Y-data"

in fisheries, apart from CKMR [and tagging]

Tedia #1: adult length but not age

We just calculated $\mathbb{P}\left[\text{xxP}|b_{\text{Julie}}a_{\text{Adam}}\ell_{\text{Adam}}y_{\text{Adam}}\right]$

But don't know a_{Adam}

So we need to work out $\mathbb{P}\left[\text{xxP}|b_{\text{J}}\ell_{\text{A}}y_{\text{A}}\right]$

$$= \sum_{a^*} \mathbb{P}\left[\text{xxP}|b_{\text{J}}, a_{\text{A}} = a^*, \ell_{\text{A}}y_{\text{A}}\right] \times \mathbb{P}\left[a^*|\ell_A y_A\right]$$

... so we need $\mathbb{P}\left[a|\ell y\right]$

Direct A@L samples? IMO the only *good* "Y-data"

in fisheries, apart from CKMR [and tagging]

Or from pop dyn & growth: $\mathbb{P}\left[a|\ell y\right] = \dfrac{\mathbb{P}\left[\ell|a\ (y\ \text{irrel..?})\right]\mathbb{P}\left[a\right]}{\sum_{a'} \text{numerator}}$

Tedia #1: adult length but not age

We just calculated $\mathbb{P}\left[\text{xxP}|b_{\text{Julie}}a_{\text{Adam}}\ell_{\text{Adam}}y_{\text{Adam}}\right]$

But don't know a_{Adam}

So we need to work out $\mathbb{P}\left[\text{xxP}|b_{\text{J}}\ell_{\text{A}}y_{\text{A}}\right]$

$$= \sum_{a^*} \mathbb{P}\left[\text{xxP}|b_{\text{J}}, a_{\text{A}} = a^*, \ell_{\text{A}}y_{\text{A}}\right] \times \mathbb{P}\left[a^*|\ell_A y_A\right]$$

... so we need $\mathbb{P}\left[a|\ell y\right]$

Direct A@L samples? IMO the only *good* "Y-data"

in fisheries, apart from CKMR [and tagging]

Or from pop dyn & growth: $\mathbb{P}\left[a|\ell y\right] = \dfrac{\mathbb{P}\left[\ell|a\ (y\ \text{irrel..?})\right]\mathbb{P}\left[a\right]}{\sum_{a'} \text{numerator}}$

My tip: fit the above to A@L data as well as CKMR

can estimate "growth curve" E[L|A] and V[...] directly

Tedia #2: the plus-group

Assume no growth & equal Z from A_+-1 onwards

Problem: back-projecting age/size for POP comps...

Can't ignore plus-group for TRO

If age known perfectly, could exclude plus-group adults...

... but not if imperfect age, or only length

3. Back-project from (~5) quantiles of that distro

Tedia #2: the plus-group

Assume no growth & equal Z from A_+-1 onwards

Problem: back-projecting age/size for POP comps...

Can't ignore plus-group for TRO

If age known perfectly, could exclude plus-group adults...

... but not if imperfect age, or only length

My preferred solution:

`. Keep track of $\bar{A}_+$ via

$$\bar{A}_{+,t+1} = \frac{N_{+t}\left(\bar{A}_{+,t}+1\right) + N_{A_+-1,t}A_+}{N_+ + N_{A_+-1,t}}$$

2. Assume geometric distro of age *within* plus-group, matching $\bar{A}_+$

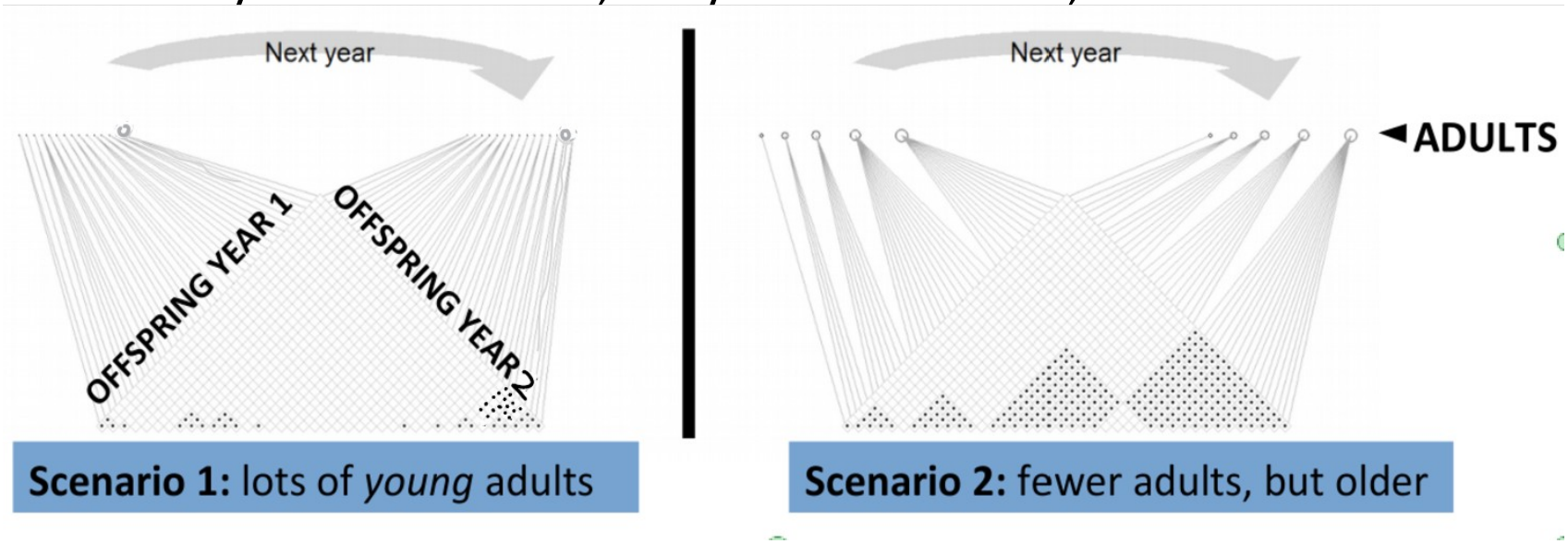
3. Back-project from (~5) quantiles of that distro

HSPs & POPs: filling the holes

These two populations have *same* TRO... 100 offspring each per year

POP rates given adult size:

~ 1/100 for *smallest*; ~8/100 for *mid-size*;

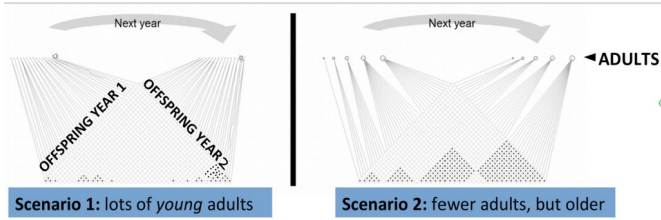


... but *very different* HSP rates !

On RHS, #HSP => "average adult must be largeish"

... thx2 *quadratic* fec term in HSPs

Nequiv again

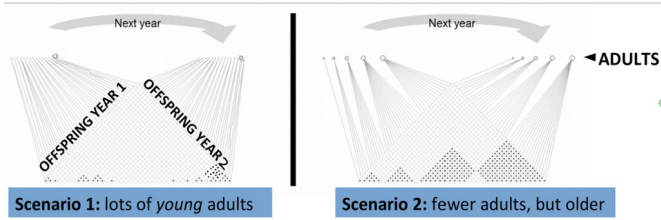


All POP probs are basically of this form:

A-specific stuff
---------- * (time-gap stuff)
TRO of J-likes

and HSP probs are like
weighted *POP probs*

Nequiv again



All POP probs are basically of this form:

A-specific stuff
---------- * (time-gap stuff)
TRO of J-likes

and HSP probs are like weighted *POP probs*

Units of TRO are up to us!

So, after allowing for time-gap stuff the *observed* rate for that AJ-category tells us:

numerical abundance Nequiv of parents of *all* J-likes, *if* all parents A-like

HSPs: Nequiv of J-like's parents if all were *average parents*;

NB average *parent* ≠ average *adult* except in "mammals"